Physics 365 or 369

Laboratory 1

Title:	Determination of the density of wood and steel.

Name	Author's Name
UCID	Author's UCID

Name	Author's Name
UCID	Author's UCID

Name	Author's Name
UCID	Author's UCID

Name	Author's Name
UCID	Author's UCID

Section	B01 – B16	Group	
Teaching Assistant		TA's Name	

Date	
Room	ST 36 (PHYS 365) or ST 38 (PHYS 369)

Determination of the Density of Wood and Steel

Objective

In 2-3 sentences state the purpose of the experiment and the goals you want to accomplish.

- *"to determine…"*
- *"to calculate…"*
- *"to compare…"*
- *"to investigate…"*

The goal of this experiment is determine the best estimate of the density, ρ, of a wooden block and a metal sphere at room temperature and to compare the calculated values within their estimated uncertainties to the expected values of density for few different types of wood (and to determine which type of wood is the wooden block made of) and steel.

Theory

*Include **important** theory that is relevant to your experiment. Quote the equations you will use. Make sure to reference the sources.*

A crucial part of any scientific experiment or engineering project is understanding and quantifying the amount of uncertainty in a measurement. This uncertainty is a critical part of determining whether an experiment agrees with a theoretical prediction, or whether it does not. The "*Bureau International des Poids et Mesures*" has produced a document called the *"Guide to Uncertainty in Measurement*" which details at length the methods for uncertainty calculations [1]. This document helps researchers account for calibration uncertainties, as well as systematic effects in a standardized manner. Today we will investigate these methods, while exploring the definition and implementation of both Type A and Type B uncertainties by attempting to estimate the density of several objects.

The first thing to consider when trying to account for the uncertainty in any given measurement is to determine the uncertainty type encountered. Type A uncertainty is derived through repeated measurements of the same value or statistical methods, as even under identical conditions there is a limit to the repeatability and reproducibility of any measurement. If a measurement of a quantity *x* is conducted *n* times, the value itself is determined by calculating the arithmetic mean, $\bar{x}$, of the quantity:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{Eq. (1)}$$

The most common method to quantify this uncertainty is calculating the standard uncertainty of the mean:

$$u(x_i) = \frac{s(x_i)}{\sqrt{n}} \qquad \text{Eq. (2)}$$

where $s(x_i)$ is the standard deviation given by:

$$s(x_i) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \qquad \text{Eq. (3)}$$

Generally speaking, one can expect that the measurement error is less than the standard uncertainty about 68% of the time.

Type B uncertainty, on the other hand, is not derived through statistical methods. This uncertainty type encompasses calibration uncertainties, and all other less rigorously defined uncertainty types. Since there is no strict method for determining these uncertainties, a level of honesty is required, as is a thorough understanding of the experimental procedures. Digital quantization can be classified as a Type B uncertainty, as are uncertainties due to rulers and analog meters. Any measurement instrument has an uncertainty associated with the last digit, so in the absence of a calibration report, it is assumed that the uncertainty with such devices is half of the smallest digit they are able to report.

Often times experimental procedure requires the combination of Type A and Type B uncertainties. More commonly, we are using measured quantities to infer a different value. In the case of this experiment we will use the mass and dimensions of various objects to estimate their mass density. In order to estimate the uncertainty in our calculated value, the uncertainty of each measurement must be considered and propagated. In the case, for example, of the density of a sphere (ρ_s) we use the diameter (D) and mass (m), to calculate the density. In this case the uncertainty is calculated using the following equation:

$$u(\rho_s) = \sqrt{\left(\frac{\partial \rho_s}{\partial D}\right)^2 u^2(D) + \left(\frac{\partial \rho_s}{\partial m}\right)^2 u^2(m)} \qquad \text{Eq. (4)}$$

Consider that:

$$\rho_s = \frac{6m}{\pi D^3} \qquad \text{Eq. (5)}$$

Then the uncertainty is given by:

$$u(\rho_s) = \sqrt{\left(\frac{-18m}{\pi D^4}\right)^2 u^2(D) + \left(\frac{6}{\pi D^3}\right)^2 u^2(m)} \qquad \text{Eq. (6)}$$

Experimental Methods

Briefly describe materials and equipment you will use. Feel free to include sketches/pictures of the experimental set up if necessary.

We begin by conducting repeated measurements of the dimensions of a wooden block and a steel sphere. These measurements are averaged to get an experimental value, and each mass is weighed using a digital scale. These measurements are taken in conjunction with the density equation to estimate the density of each item, as well as the associated uncertainty of each measurement.

Data

Your data has to be consistent with the raw data you have submitted via D2L before leaving the lab. Use tables to organize your data.

Measurements of a Wooden Block

Mass (g)	**Uncertainties**
12.2	u(mass) = 0.1 g
Side Length, l, (mm)	
30.00	u(l) = 0.1 mm
29.84	
29.50	
29.44	
30.00	
29.84	
Average Length	**29.77 mm**

Table (1)

Measurements of a Steel Sphere

Mass (g)	**Uncertainties**
67.2	u(mass) = 0.1 g
Diameter, D, (mm)	
25.34	u(D) = 0.1 mm
35.38	
25.36	
25.40	
25.32	
25.35	
Average Diameter:	**25.36 mm**

Table (2)

Analysis

Calculate all values and estimate their uncertainties.

The average density of a perfect cube of a given mass, *m*, and volume, *V*, with a side length of *l*, is given by:

$$\bar{\rho}_{cube} = \frac{m}{V} = \frac{m}{l^3} \qquad \text{Eq. (7)}$$

In our case this becomes:

$$\bar{\rho}_{cube} = \frac{12.2\ g}{(29.77\ mm)^3} = 4.6 \times 10^{-4}\ {}^{g}/_{mm^3} = 0.46\ {}^{g}/_{cm^3}$$

The uncertainty for the density of the cube is then given by:

$$u(\rho_{cube}) = \sqrt{\left(\frac{\partial \rho_{cube}}{\partial l}\right)^2 u^2(l) + \left(\frac{\partial \rho_s}{\partial m}\right)^2 u^2(m)}$$

$$u(\rho_{cube}) = \sqrt{\left(\frac{-3m}{l^4}\right)^2 u^2(l) + \left(\frac{1}{l^3}\right)^2 u^2(m)} = \sqrt{\left(\frac{-3m}{l^4}\right)^2 u^2(l) + \left(\frac{1}{l^3}\right)^2 u^2(m)}$$

$$u(\rho_{cube}) = \sqrt{\left(2.171 \times 10^{-9}\ {}^{g^2}/_{mm^8}\right)(0.01\ mm^2) + \left(1.437 \times 10^{-9}\ {}^{}/_{mm^6}\right)(0.01\ g^2)}$$

$$u(\rho_{cube}) = 6.01 \times 10^{-6}\ {}^{g}/_{mm^3} = 0.01\ {}^{g}/_{cm^3}$$

Therefore, our wooden block has a density of:

$$\rho_{cube} = 0.46\ {}^{g}/_{cm^3}\ ; \quad u(\rho_{cube}) = 0.01\ {}^{g}/_{cm^3}$$

We use a similar method for the steel sphere. As shown in the Theory section, the average density of a sphere of diameter *D*, is given by:

$$\rho_s = \frac{6m}{\pi D^3} = \frac{6\ (67.2\ g)}{\pi (25.36\ mm)^3} = 0.00787\ {}^{g}/_{mm^3} = 7.87\ {}^{g}/_{cm^3}$$

The uncertainty is then given by:

$$u(\rho_s) = \sqrt{\left(\frac{\partial \rho_s}{\partial D}\right)^2 u^2(D) + \left(\frac{\partial \rho_s}{\partial m}\right)^2 u^2(m)}$$

$$u(\rho_s) = \sqrt{\left(\frac{-18m}{\pi D^4}\right)^2 u^2(D) + \left(\frac{6}{\pi D^3}\right)^2 u^2(m)}$$

$$u(\rho_s) = 9.4 \times 10^{-5}\, {}^{g}/_{mm^3} = 0.1\, {}^{g}/_{cm^3}$$

Therefore, our steel sphere has a density of:

$$\rho_{sphere} = 7.9\ {}^{g}/_{cm^3}\ ; \quad u(\rho_{sphere}) = 0.1\ {}^{g}/_{cm^3}$$

Discussion

Compare the obtained values to the expected ones using the t-test. Comment on the agreement and on possible sources of errors (both random and systematic). Make sure the errors are reasonable given your results.

Wooden Block

In order to discuss our results in the context of accepted values, we introduce a table detailing the densities of various types of wood.

Wood Type	Density (g/cm^3)	Density Uncertainty (g/cm^3)
Bamboo	0.35	0.05
Oak	0.75	0.15
White Pine	0.43	0.08

Table (3)

The density value obtained for our wooden block was 0.46 g/cm^3, with an uncertainty u(ρ) = 0.01 g/cm^3. Upon first glance, it is possible the wood block was made of white pine, though the uncertainty in the measurement suggests it could also possibly be bamboo or oak. To be sure, we will conduct the Student's t-test on each of these values. The test is conducted using the following equation, in this case, calculating for white pine.

$$|t| = \left| \frac{\rho_{experimental} - \rho_{accepted}}{\sqrt{u(\rho_{experimental})^2 + u(\rho_{accepted})^2}} \right| = \left| \frac{0.43 - 0.43}{\sqrt{(0.01)^1 + (0.08)^2}} \right| = 0.4 \qquad \text{Eq. (8)}$$

The results are as follows [2]:

Wood Type	|t|
Bamboo	2.2
Oak	1.9
White Pine	0.4

Table (4)

We can say then, that the wooden block is most likely made of white pine. The *t* value less than 2 means that our results agree within uncertainty, while the other wood species have a much larger *t* value, suggesting less agreement.

There are a few systematic errors that were introduced to this calculation, the first and largest being that we assumed the block is of constant density for the sake of calculations. We know for a fact this is not true, as the block had letters cutout into the side. Furthermore, we noticed a slight discrepancy between the lengths obtained depending on which side of the block was measured. This suggests that the block was not a perfect cube, as assumed in the derivations. Regardless of these factors, we were able to obtain an experimental value which, together with its uncertainty, agrees with the literature value for white pine.

Steel Sphere

Depending on the chemical composition, steel is expected to have a density ranging from (7.75-8.05) g/cm^3 [3]. The value we found experimentally for our steel ball is 7.9 g/cm^3, with an uncertainty of u(ρ) = 0.1 g/cm^3. The mean value of the accepted density range is 7.9 g/cm^3, which is in complete agreement with the measured value, or would give a t value of zero.

While this is an extremely unlikely experimental result, it is likely due to the highly precise and repeated measurements conducted on the sphere. Calipers were used for measurement, which allow for measurement within fractions of a millimeter. We had multiple group members conduct this measurement, which likely means systematic biases may have been lessened or perhaps could have counteracted one another.

It is quite possible that sources of error were still present within this portion of our experiment. We noticed a slight depression on one side of the steel ball, suggesting it may have been dropped in the past. This would lead to an imperfect sphere, though we tried to avoid that part of the sphere to lessen this effect. We have verified the density equation given in Equation 4 through this experiment.

Conclusion

Report your values again and comment on their agreement with accepted values

We measured the dimensions and masses of a wooden block and a steel sphere in order to test the density equation:

$$\bar{\rho} = \frac{m}{V}$$

We found an experimental value for the wooden block to be 0.46 g/cm^3, with an uncertainty u(ρ) = 0.01 g/cm^3. This is in agreement with the accepted density of white pine, 0.4 g/cm^3, giving a *t*-test value of $|t|$=0.4. We then measured the density of a steel sphere, giving an experimental value of is 7.9 g/cm^3, with an uncertainty of u(ρ) = 0.1 g/cm^3. This is in agreement with the accepted value for the density of steel, (7.75-8.05) g/cm^3, giving a t-test value of 0.

We found reasonable agreement with the accepted values of both of our measured densities within their uncertainties. In the experiment both types of uncertainties A and B were estimated.

References

[1] Joint Committee for Guides in Meteorology. JCM 100*: Evaluation of measurement data – guide to the expression of uncertainty in measurement*. Technical report, JCGM, 2008. [www.bipm.org/en/publications/guides/gum.html]

[2] Engineering ToolBox, (2004). *Densities of Wood Species*. Available at: https://www.engineeringtoolbox.com/wood-density-d_40.html [Accessed 07 09 2018].

[3] Haynes, W. M. (2009). *CRC handbook of chemistry and physics: A ready-reference book of chemical and physical data*. Boca Raton: CRC Press.